NAME : - U.PRANEETH ; ROLL NO: 2520030188

SEC :- 03

WEEK :- 1

Step 1: Create Database

```sql
DROP DATABASE IF EXISTS student_db3;

CREATE DATABASE student_db3;

USE student_db3;
```

OUTPUT:-

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	11:13:43	DROP DATABASE IF EXISTS student_db3	1 row(s) affected	0.032 sec
2	11:13:46	CREATE DATABASE student_db3	1 row(s) affected	0.047 sec
3	11:13:48	USE student_db3	0 row(s) affected	0.000 sec

STEP 2:- Create Table

```sql
DROP TABLE IF EXISTS student_marks11;

CREATE TABLE student_marks11 (
    roll_no INT PRIMARY KEY,
    name VARCHAR(50),
    subject VARCHAR(50),
    marks DECIMAL(5,2)
);
```

OUTPUT:-

Action Output

#	Time	Action	Message	Duration / Fetch
1	11:25:49	DROP TABLE IF EXISTS student_marks11	0 row(s) affected	0.031 sec
2	11:25:52	CREATE TABLE student_marks11 (roll_no INT PRIMARY KEY, name VARCHAR(50), subject VARCHAR...	0 row(s) affected	0.031 sec
3	11:25:54	CREATE TABLE student_marks11 (roll_no INT PRIMARY KEY, name VARCHAR(50), subject VARCHAR...	Error Code: 1050. Table 'student_marks11' already exists	0.000 sec
4	11:25:56	CREATE TABLE student_marks11 (roll_no INT PRIMARY KEY, name VARCHAR(50), subject VARCHAR...	Error Code: 1050. Table 'student_marks11' already exists	0.000 sec

---

Step 3: Insert Student Records

```sql
INSERT INTO student_marks11 (roll_no, name, subject, marks) VALUES
(1,'Ravi','Math',85.50),
(2,'Sita','Math',92.75),
(3,'Anil','Math',78.40),
(4,'Priya','Math',88.90),
(5,'Vijay','Math',80.25),
(6,'Subbusir','AWS',98.50),
(7,'Dwaraka','DBMS',95.75),
(8,'Ranjani','English',97.40),
(9,'Kavitha','AWS',99.90),
(10,'Seetha','Azure',82.25);
```

OUTPUT:-

```sql
DROP TABLE IF EXISTS student_marks11;

CREATE TABLE  student_marks11 (
    roll_no INT PRIMARY KEY,
    name VARCHAR(50),
    subject VARCHAR(50),
    marks DECIMAL(5,2),
);

INSERT INTO student_marks11 (roll_no, name, subject, marks) VALUES
(1,'Ravi','Math',85.50),
(2,'Sita','Math',92.75),
(3,'Anil','Math',78.40),
(4,'Priya','Math',88.90),
(5,'Vijay','Math',80.25),
(6,'Subbusir','AWS',98.50),
(7,'Dwaraka','DBMS',95.75),
(8,'Ranjani','English',97.40),
(9,'Kavitha','AWS',99.90),
(10,'Seetha','Azure',82.25);

SELECT * FROM student_marks11;
SELECT COUNT(*) AS total_students FROM student_marks11;
```

Action Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	11:55:21	DROP TABLE IF EXISTS student_marks11	0 row(s) affected	0.016 sec
2	11:55:21	CREATE TABLE student_marks11 (roll_no INT PRIMARY KEY, name VARCHAR(50), subject VAR...	0 row(s) affected	0.031 sec
3	11:55:21	INSERT INTO student_marks11 (roll_no, name, subject, marks) VALUES (1,'Ravi','Math',85.50), (2,...	10 row(s) affected	0.047 sec
4	11:55:21	SELECT * FROM student_marks11	10 row(s) returned	0.000 sec / 0.000 sec
5	11:55:21	SELECT COUNT(*) AS total_students FROM student_marks11	1 row(s) returned	0.000 sec / 0.000 sec

Result Grid | Filter Rows: | Edit: | Export: | Wrap Cell Content:

roll_no	name	subject	marks
1	Ravi	Math	85.50
2	Sita	Math	92.75
3	Anil	Math	78.40
4	Priya	Math	88.90
5	Vijay	Math	80.25
6	Subbusir	AWS	98.50
7	Dwaraka	DBMS	95.75
8	Ranjani	English	97.40
9	Kavitha	AWS	99.90
10	Seetha	Azure	82.25

Result Grid | Filter Rows: | Edit: | Export: | Wrap Cell Content:

total_students
10

Step 4: Display All Records

```sql
SELECT * FROM student_marks11;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

roll_no	name	subject	marks
1	Ravi	Math	85.50
2	Sita	Math	92.75
3	Anil	Math	90.00
4	Priya	Math	88.90
6	Subbusir	AWS	98.50
7	Dwaraka	DBMS	95.75
8	Ranjani	English	97.40
9	Kavitha	AWS	99.90
10	Seetha	Azure	82.25

Result Grid

Form Editor

Field Types

student_marks11 27 × | Read Only | Context Help | Snippets

OUTPUT:-

student_marks11 27 × Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	11:42:44	SELECT * FROM student_marks11 LIMIT 0, 1000	9 row(s) returned	0.000 sec / 0.000 sec

Step 5: Count Students

```sql
SELECT COUNT(*) AS total_students
FROM student_marks11;
```

OUTPUT:-

Action Output

#	Time	Action	Message	Duration / Fetch
1	11:44:51	SELECT COUNT(*) AS total_students FROM student_marks11 LIMIT 0, 1000	1 row(s) returned	0.015 sec / 0.000 sec
2	11:44:55	SELECT COUNT(*) AS total_students FROM student_marks11 LIMIT 0, 1000	1 row(s) returned	0.016 sec / 0.000 sec

Step 6: Update Records

```sql
UPDATE student_marks11
SET marks = 90.00
WHERE roll_no = 3;

UPDATE student_marks11
SET subject = 'Remedial Math'
WHERE marks < 80;
```

OUTPUT:-

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	11:48:25	SELECT COUNT(*) AS total_students FROM student_marks11 LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
2	11:48:29	UPDATE student_marks11 SET marks = 90.00 WHERE roll_no = 3	0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0	0.015 sec
3	11:48:30	UPDATE student_marks11 SET marks = 90.00 WHERE roll_no = 3	0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0	0.000 sec
4	11:48:32	UPDATE student_marks11 SET marks = 90.00 WHERE roll_no = 3	0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0	0.000 sec

Step 7: Delete Records

```sql
SELECT * FROM student_marks11;

DELETE FROM student_marks11
WHERE roll_no = 5;

DELETE FROM student_marks11
WHERE marks < 75;
```

OUTPUT:-

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	11:51:56	DELETE FROM student_marks11 WHERE roll_no = 5	0 row(s) affected	0.000 sec
2	11:51:58	DELETE FROM student_marks11 WHERE roll_no = 5	0 row(s) affected	0.000 sec
3	11:52:01	DELETE FROM student_marks11 WHERE marks < 75	0 row(s) affected	0.000 sec
4	11:52:03	DELETE FROM student_marks11 WHERE marks < 75	0 row(s) affected	0.000 sec

---

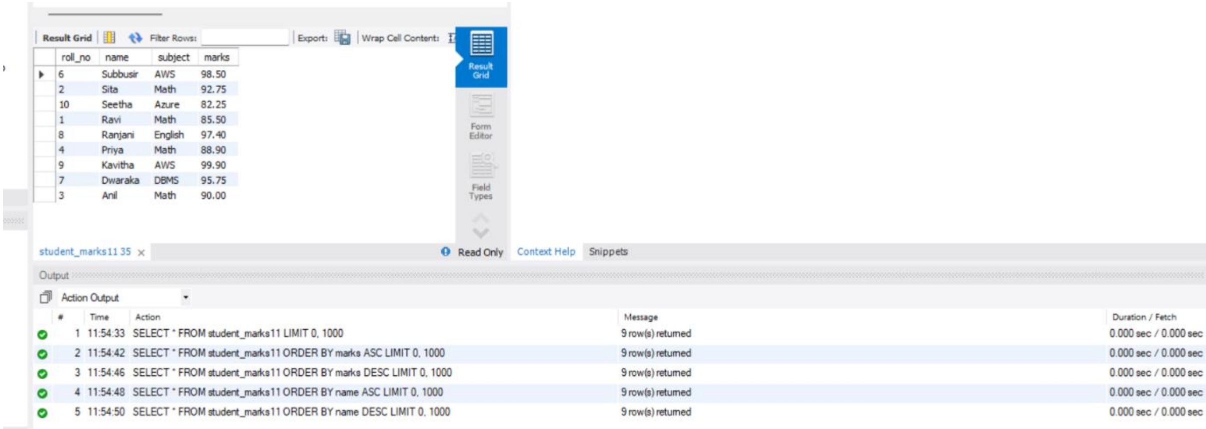
Step 8: Sort Records

```sql
SELECT * FROM student_marks11;

SELECT * FROM student_marks11 ORDER BY marks ASC;
SELECT * FROM student_marks11 ORDER BY marks DESC;
SELECT * FROM student_marks11 ORDER BY name ASC;
SELECT * FROM student_marks11 ORDER BY name DESC;

SELECT subject, SUM(marks) AS total_marks
FROM student_marks11
GROUP BY subject;
```

OUTPUT:-

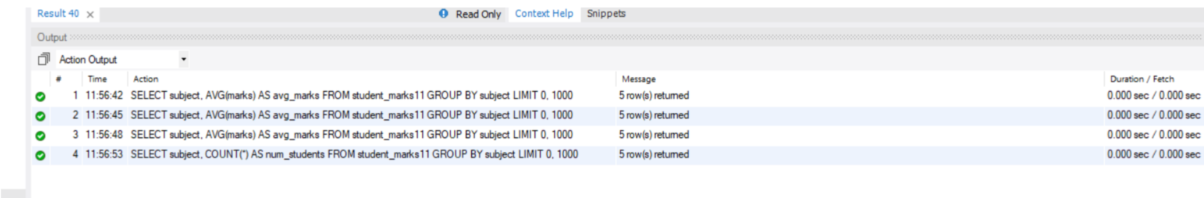


Step 9: Aggregate Functions

```
SELECT subject, AVG(marks) AS avg_marks
FROM student_marks11
GROUP BY subject;

SELECT subject, COUNT(*) AS num_students
FROM student_marks11
GROUP BY subject;
```

OUTPUT:-



Step 10: HAVING Clause

```sql
SELECT subject, AVG(marks) AS avg_marks
FROM student_marks11
GROUP BY subject
HAVING AVG(marks) > 90;

SELECT subject, COUNT(*) AS num_students
FROM student_marks11
GROUP BY subject
HAVING COUNT(*) > 1;
```

OUTPUT:-